

AI/ML Internship – Day 63 Notes & Tasks

Module 8: AI Chatbot using Machine Learning Concepts (Intent Classification)

Part 1: Recap of Day 62

Yesterday we learned:

-  NLP using NLTK
-  Tokenization
-  Stop Word Removal
-  Stemming
-  Lemmatization
-  Building a simple NLP chatbot

Today we will learn how an **AI chatbot understands the user's intention using Machine Learning**.

Part 2: What is an AI Chatbot?

Definition

An AI Chatbot is a chatbot that uses **Artificial Intelligence, Machine Learning, and Natural Language Processing** to understand user messages and generate intelligent responses.

Unlike rule-based chatbots, AI chatbots can understand different ways of asking the same question.

Simple Definition

An AI chatbot learns patterns from data instead of depending only on predefined rules.

Examples

- ChatGPT
- Google Gemini
- Microsoft Copilot

- Siri
- Alexa

✦ Part 3: Rule-Based Chatbot vs AI Chatbot

Rule-Based Chatbot

Uses fixed rules

Exact matching

Cannot improve automatically

Best for FAQs

AI Chatbot

Learns from data

Understands different sentence patterns

Can improve with better training data

Best for conversations

✦ Part 4: What is Intent Classification?

Definition

Intent Classification is the process of identifying the user's purpose from a message.

It is one of the most important tasks in chatbot development.

Example

User:

I forgot my password.

Intent:

Password Reset

User:

Where is my order?

Intent:

Track Order

User:

Can you tell me my account balance?

Intent:

Balance Enquiry

Part 5: How AI Chatbots Learn

An AI chatbot learns from **training data**.

Training Data contains:

- User Message
- Correct Intent

Example:

User Message	Intent
Hello	Greeting
Hi	Greeting
Good Morning	Greeting
Track my order	Order Tracking
Where is my parcel?	Order Tracking
Check my order status	Order Tracking

The chatbot studies these examples and learns that all these sentences have similar meanings.

Part 6: Chatbot Training Workflow

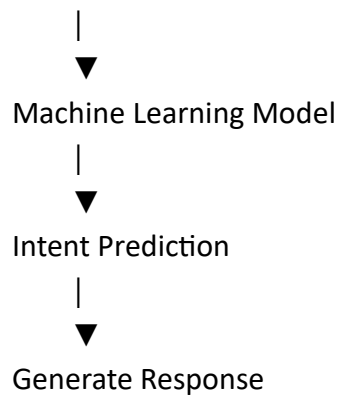
Training Dataset



Text Preprocessing



Feature Extraction



✚ Part 7: Feature Extraction

Definition

Computers cannot understand text directly.

So text must be converted into numbers.

This process is called **Feature Extraction**.

Example

Sentence:

I want to book a ticket



Convert into numbers using techniques like:

- Bag of Words (BoW)
- TF-IDF
- Word Embeddings



Machine Learning model processes the numbers.

✚ Part 8: Simple Intent Dataset

User Message	Intent
Hi	Greeting

User Message	Intent
Hello	Greeting
Good Evening	Greeting
Bye	Goodbye
See you	Goodbye
Track my order	Order Tracking
Where is my order?	Order Tracking
Cancel my order	Order Cancellation
Refund my payment	Refund

🔥 Part 9: Simple AI Chatbot (Intent Prediction)

```
training_data = {
    "hello": "Greeting",
    "hi": "Greeting",
    "bye": "Goodbye",
    "track order": "Order Tracking",
    "refund": "Refund"
}

user = input("You: ").lower()

for text, intent in training_data.items():
    if text in user:
        print("Predicted Intent:", intent)
        break
else:
    print("Intent not recognized.")
```

Note: This is a simplified simulation. Real AI chatbots use trained machine learning models instead of manually checking keywords.

🔥 Part 10: Why Do We Need Training Data?

Without training data:

✗ Chatbot cannot learn.

With training data:

✓ Chatbot recognizes similar user messages.

The quality of the chatbot depends heavily on the quality and diversity of the training data.

✦ Part 11: Importance of Large Datasets

Suppose the chatbot is trained only with:

Track my order

Now the user asks:

Where is my package?

The chatbot may fail because it has never seen this sentence.

If the dataset includes many variations:

- Track my order
- Where is my package?
- Order status
- Has my parcel shipped?

The chatbot performs much better.

✦ Part 12: Real-World Applications

Banking

- Balance enquiry
 - Loan enquiry
 - Credit card support
-

Healthcare

- Appointment booking
 - Medicine information
-

Education

- Admission enquiry
 - Fee details
-

E-commerce

- Order tracking
 - Returns
 - Refunds
-

✦ Part 13: Advantages of AI Chatbots

- ✓ Understand natural language
 - ✓ Learn from training data
 - ✓ Handle different sentence structures
 - ✓ Better customer experience
 - ✓ Scalable
-

✦ Part 14: Limitations

- ✗ Requires large datasets
 - ✗ Needs proper training
 - ✗ Can make incorrect predictions
 - ✗ Performance depends on data quality
-

✦ Part 15: Theory Questions

What is an AI Chatbot?

An AI chatbot uses Artificial Intelligence, Machine Learning, and NLP to understand and respond to user messages.

What is Intent Classification?

Intent Classification identifies the purpose behind a user's message.

Why is training data important?

Training data helps the chatbot learn different ways users ask similar questions.

What is Feature Extraction?

Feature Extraction converts text into numerical values so that machine learning models can process it.

Why do AI chatbots perform better than rule-based chatbots?

Because they can understand different sentence patterns instead of relying only on exact matches.

Part 16: Interview Questions

What is intent classification?

Intent classification is the process of predicting the user's goal from a message.

Why is feature extraction necessary?

Because machine learning algorithms work with numerical data, not raw text.

What factors affect chatbot accuracy?

- Dataset quality
 - Dataset size
 - Proper preprocessing
 - Model selection
-

Why is a large dataset important?

A larger dataset provides more examples, helping the chatbot generalize better to new user inputs.

Name three techniques used for feature extraction.

- Bag of Words (BoW)
 - TF-IDF
 - Word Embeddings
-

Tasks for Day 63

Theory

1. Define an AI chatbot.
 2. Explain intent classification.
 3. Why is training data important?
 4. What is feature extraction?
 5. Compare rule-based and AI chatbots.
-

Practical Task 1

Create a table containing **15 user messages** and their corresponding intents.

Practical Task 2

Identify the intent for these messages:

- "Where is my order?"
 - "Cancel my booking."
 - "Hi!"
 - "Refund my money."
-

Practical Task 3

Create your own **college enquiry intent dataset** with at least **20 training examples**.

Practical Task 4

Write a Python program that predicts an intent using a simple dictionary lookup.

Practical Task 5

Draw the chatbot training workflow diagram.